

Pietro Girotto

MACHINE LEARNING ENGINEER

📞 (+39) 3774120247 | ✉️ pietrogirotto00@gmail.com | 🏠 pi.gp-apps.org | 📱 gp-1108 | 🌐 pietro-girotto

Work Experience

Swisscom

Bern, Switzerland

MACHINE LEARNING ENGINEER IIP

02/2025 - Present

- Gained full familiarity with **ML/Dev Ops pipelines**: development to deployment via **CI/CD** pipelines to **AWS Kubernetes** cluster.
- Improved inference times for in-house deployed **Chatbot** application, making it **x15 faster** while reducing querying costs by 80%.
- Full-stack development and deployment of an **AI Agent from scratch** tool used by the whole department (**FastAPI** and **NextJS**)
- Custom-built machine learning advanced evaluation pipelines.

Uber

Krakow, Poland

AUTOMATION SPECIALIST INTERN

02/2024 - 05/2024

- Executed end-to-end problem-solving projects mainly in **Python** and **SQL**, from requirements gathering to post-deployment support.
- Discovered and dealt with bugs in legacy code that were affecting as much as **40% of the user base**.
- Achieved **exclusively maximum ratings** in stakeholders' satisfaction, delivery time and communication.

Education

University of Padua - Artificial Intelligence

Padua, Italy

MSC IN COMPUTER ENGINEERING

09/2022 - 03/2025

- Final grade: 110/110 with honors, **GPA of 4.0**.
- Experimental thesis on **LLM fine-tuning** and **reinforcement learning** alignment, training models on a multi-gpu bare metal cluster.
- Hands on experience in **Machine Learning** frameworks and concepts. Fine-tuning of **Deep learning** models for domain-specific (**Computer Vision**, **Natural Language Processing**) purposes in **PyTorch**.
- Further refined my programming skills with **C++**, Python and **Java** projects. Gained knowledge and experience in Big Data frameworks such as Apache Spark (Java) and image manipulation (OpenCV).

University of Padua

Padua, Italy

BSC IN COMPUTER ENGINEERING

09/2019 - 07/2022

- Final grade: 110/110 **with honors**, GPA 4.0 and **top 3%** of my course of study.
- **Experimental thesis** about Integer Linear Programming exploring alternatives to SotA models.
- Solid foundation in computer science with hands-on experience in full-stack **web development** (React, JS, Express, NodeJS, Google Cloud) and more traditional languages (C++, Java, C).

Projects

Educational AI with Llama 3.1 & DPO

PYTHON, LLM, HUGGING FACE, PYTORCH, NATURAL LANGUAGE PROCESSING (NLP)

Developed a cutting-edge LLM training pipeline, combining automated dataset generation, parameter-efficient fine-tuning, and Direct Preference Optimization to deliver next-gen educational AI. [Code]

Occluded people image detector

PYTHON, FLASK, C++, HTML5, CSS, PYTORCH, COMPUTER VISION (CV)

Fine-tuned a deep learning model to accurately segment occluded bodies on a dataset of 40K images using distributed computing, achieving over 97% precision on the test set. [Code | Demo | Report]

AI Playing Snake

PYTHON, FLASK, HTML5, CSS, TENSORFLOW, REINFORCEMENT LEARNING (RL)

Implemented a Deep Reinforcement Learning Agent from scratch to play the Snake Game. By training it over 1M samples, the agent's performances increased by 20% with respect to the baseline. [Code | Demo | Report]

LLM Code Evaluation

PYTHON, LLM, NATURAL LANGUAGE PROCESSING (NLP)

Developed an evaluation framework for LLM code completion to assess model performance through advanced metrics. [Code | Report]

Volunteering

Croce Verde Padova

Padua, Italy

TEAMWORK, COMMUNICATION, LEADERSHIP, WORKING UNDER PRESSURE

02/2022 - 02/2025

Taken part consistently in emergency ambulance service in my town, dealing with high stress environments and situations.